

Module 5: Basic Computer and Information Theory — Answer Key

Prequiz

1.

$$13 = 8 + 4 + 1 \Rightarrow \boxed{1101_2}.$$

Also,

$$10110_2 = 16 + 4 + 2 = \boxed{22}.$$

2.

$$0.101_2 = \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = \boxed{0.625}.$$

3/16 terminates because $16 = 2^4$. 3/10 does not terminate because the reduced denominator has a factor of 5.

3. Data 1011 has three 1s, so parity bit $\boxed{1}$ makes four. Packet 10110 has three 1s, so an even-parity receiver detects an error.

4.

$$\text{HHHHHTTTTT} \rightarrow \boxed{5\text{H}5\text{T}}.$$

The string XYZXYZXYZ has almost no consecutive repetition, so run-length encoding does not shorten it.

5.

$$\frac{1}{32} = 2^{-5} \Rightarrow I = \boxed{5 \text{ bits}}.$$

Matching quiz

1.

$$22 = 16 + 4 + 2 \Rightarrow \boxed{10110_2}, \quad 10001_2 = 16 + 1 = \boxed{17}.$$

2.

$$0.011_2 = \frac{1}{4} + \frac{1}{8} = \boxed{0.375}.$$

5/8 terminates; 1/5 does not.

3. Data 0111 has three 1s, so parity bit $\boxed{1}$. Packet 11011 has four 1s, so it passes the even-parity check.

4. MMMMMMMMMM $\rightarrow$ $\boxed{10\text{M}}$, which is shorter than writing ten M's.

5. Probability = $\boxed{1/128}$. Since $1/128 = 2^{-7}$, information = $\boxed{7}$ bits.